

Research part for the proposal of a Scientific Event

Title: Biodiversity Dynamics in Coevolutionary Meta-ecosystems (COMETA)

Carlos J. Melián

1. Scientific aspects

1.1 Introduction of the research subject

Coevolution, the trait change driven by reciprocal selection within and among interacting species, has been invoked as a primary mechanism generating much of Earth's biodiversity (Althoff et al., 2014, Hembry et al., 2014). Two main theories have been proposed to explain the dynamics of coevolutionary diversification. The Fisherian sexual selection theory, which states that female and male traits can lead to sexual isolation and speciation (Fisher, 1930, Pomiankowski and Iwasa, 1998), and the Geographic mosaic theory of coevolution (Thompson, 2005), which suggests that reciprocal changes in traits determining species interaction produce coevolutionary diversification as interspecific interactions control reproductive isolation and speciation. Despite the appeal of such theories for explaining much of Earth's biodiversity, understanding how coevolution of intra- and inter-specific traits associated with ecological interactions alter ecosystem dynamics across heterogeneous landscapes remains elusive. Such knowledge understanding is particularly important in ecosystems with species- and interaction-rich ecological networks because the global change might trigger many species extinctions and the collapse of ecosystems (Thompson, 2014).

In this vein, To address this challenge we will join theoretical and empirical evolutionary ecologists to develop theoretical models that incorporate the empirical patterns of reciprocal trait changes will be critical to improve our predictions on biodiversity dynamics in species- and interaction-rich ecological communities. The aim of the workshop is to integrate hierarchical trait-species-space interaction data with new process-based coevolutionary metaecosystem models. Such integration might facilitate the exploration of patterns of coexistence of co-evolved species, their role in ecosystem function, and the risk of collapse of co-evolved ecosystems across different spatio-temporal scales, which is of critical importance for three reasons. First, the extinction of species caused by human-disturbances is occurring at an unprecedented rate, yet, most models predicting extinction rates often assume that species evolve independently of one another (Thompson, 2014). Second, despite recent advances in eco-evolutionary approaches, studies connecting co-evolving trait distributions within and between species at small scales to investigate the large-scale association between coevolutionary hot spots, biodiversity function and meta-ecosystem dynamics are rare (Doebeli and Ispolatov, 2017). Third, although many experiments and empirical studies have shown convergence of time scales between ecological and coevolutionary processes (Hairston et al., 2005, Fussmann et al., 2007, Schoener, 2011), there is a gap in understanding how ecological and coevolutionary dynamics in species-rich assemblages shape biodiversity and ecosystem function in a spatio-temporal context – i.e. the meta-ecosystem

(Gomulkiewicz et al., 2000, Thompson, 2013).

A meta-ecosystem is a set of local ecosystems connected by spatial flows of energy, materials and organisms across ecosystem boundaries (Loreau et al., 2003). The integration of meta-ecosystem ecology, coevolutionary processes within and between species and ecosystem functioning remains one of the main challenges for theoretical and applied ecologists. This paradigm provides a powerful tool to understand the properties that arise from the spatial coupling of local ecosystems, such as coevolutionary selection mosaics, stabilization of ecosystem processes and indirect interactions at regional scales (Gomulkiewicz et al., 2000, Gravel et al., 2010). We propose a coupled workshop and research visit aiming to integrate empirical datasets, theory and analytical methods on coevolution and meta-ecosystems to bridge the gap between coevolutionary and ecological spatio-temporal dynamics of species- and interaction-rich networks. This integration will help us to advance our understanding of biodiversity dynamics and ecosystem functioning across spatial and temporal scales.

We believe it is the right time for such a workshop. There have recently been many advances in mathematical approaches for studying coevolution (Althoff et al., 2014, Hembry et al., 2014, Doebeli and Ispolatov, 2017). However, most of these approaches are indirectly connected to biodiversity dynamics and meta-ecosystems functioning, and assume that species coevolve in isolated communities. This assumption, which is more associated to mathematical convenience than to biological and natural history knowledge, limits our understanding of the relationships among coevolution, biodiversity dynamics and ecosystem function across spatio-temporal scales in species-rich interacting systems. At the same time, there

During the workshop and research visit we will combine methods and models to incorporate co-evolving trait distributions and species interactions in meta-ecosystems (Gravel et al., 2016, DeLong and Gibert, 2016, Massol et al., 2017, Andreatzi et al., 2018, Melián et al., 2018). Such approaches will consider spatially variable biotic selection and individual migration among heterogeneous patches to generate complex selection mosaics, leading to spatial variation in species traits, and therefore to predictions of the structure of ecological networks in meta-ecosystems. Therefore, we will seek to combine coevolutionary dynamics, meta-ecosystems and interaction networks, by synthesizing spatial and trait-interaction data in species-rich ecosystems to contribute.

1.2 Scientific aims and methods of the workshop

We will focus on implementing analytic tools and testing them with empirical patterns from data sets containing many species, sites and traits (Heleno et al., 2012, Aizen et al., 2016, Frickel et al., 2016, Weinstein and Graham, 2017) (Table 1). We propose bringing together biologists skilled in theoretical ecology, evolutionary biology, and community ecology. We will focus on the following two questions about co-evolution in meta-ecosystems.

Question 1: How does reciprocal trait change influence biodiversity dynamics, species

extinctions and ecosystem function in species-rich meta-ecosystems?

We aim to integrate the effects of ecological, coevolutionary, and migration dynamics in shaping biodiversity in meta-ecosystems. We will merge the meta-ecosystem perspective (Gravel et al., 2016, Massol et al., 2017, Melián et al., 2018) into existing models of coevolutionary interaction networks (Guimarães et al., 2011, Andreazzi et al., 2017). We will implement scenarios to predict biodiversity distribution across space and time under different functional interactions (*i.e.*, *trait matching and exploitation barriers*) assuming weak and strong reciprocal selection strengths. Metacommunity and meta-ecosystem theory approaches will complement the functional scenarios to explore how spatial processes governed by dispersal (*i.e.* mass effect) or niche width (*i.e.* species sorting, patch dynamics) can alter the coevolutionary dynamics within communities by contributing to species trait variation and network structure and stability, and thus ecosystem functioning (Mouquet and Loreau, 2003, Leibold et al., 2004, Logue et al., 2011, Winegardner et al., 2012, Borthagaray et al., 2014 a,b, 2015a, Astegiano et al., 2015a, Laroche et al., 2016, Massol et al., 2017 a, b).

Question 2: How do the spatial arrangement of ecosystems and coevolutionary dynamics interact to facilitate rapid evolutionary changes in meta-ecosystems?

Most interactions between species generate reciprocal selection pressures, and the coevolutionary dynamics that emerges over time varies with the fitness consequences of the ecological interaction (*i.e.*, competitive, mutualistic, or antagonistic) and the architecture of the coevolving interactions between species (Wade 2007, Loeuille and Loreau, 2005, Doebeli 2011, Nuismer et al., 2013, Allhoff and Drossel, 2013, Althoff et al., 2014, Hembry et al., 2014). Yet, how the relationship between species traits and coevolution at local scales and the arrangement of ecosystems at the landscape scale produce rapid biodiversity changes and coevolutionary hot spots is not well understood (Wade, 2007, Doebeli, 2011, Althoff et al., 2014, Hembry et al., 2014). We will run simulations under landscapes with different spatial arrangements and migration scenarios to disentangle the interplay between coevolutionary dynamics at the local and the landscape level.

1.3 Expected results

We will work on an opinion, review or ideas-perspectives paper entitled “*Biodiversity dynamics and ecosystem functioning in coevolutionary meta-ecosystems*” for submission to *Trends in Ecology and Evolution*, *Ecology Letters* or *Biological reviews*. The two questions above will form the core of the discussions at the workshop, and they will help drafting an outline for the paper during the meeting. Prior to the meeting, attendees will provide written perspectives about how to address each of the overarching questions and the trait-species-spatial interaction data they can provide. On the first day, participants will present these ideas based on their own research programs. On subsequent days we will synthesize these perspectives into a single review/opinion and synthesis style paper integrating the datasets to the methods that we will develop and implement during the workshop and following research visit stay.

2. Organizational aspects

2.1 Milestones and program

The main milestones of the workshop will be:

1. Draft the outline for a synthesis paper about Biodiversity dynamics and ecosystem functioning in coevolutionary meta-ecosystems summarizing the main findings obtained during the workshop.
2. Discuss future directions for the field that unify short- and long-term approaches integrating modeling and data.
3. Integrate new modeling frameworks to empirical data and make them available throughout github and related open repositories to facilitate reproducibility of the analysis.
4. Develop new ideas to strengthen future grant proposals and collaborations

2.2 Venue

The workshop will be hosted at Eawag's Center for Ecology, Evolution & Biogeochemistry (CEEB), in Kastanienbaum. This is an ideal venue. All the invited guests can stay on-site in Eawag apartments, directly on the shore of Lake Lucerne. The meeting facilities are very suitable and have been used before for such workshops. We have previously used this location to host several small workshops and student courses with great success. Such events also help to raise our international profile, and, in the past, have led to repeat visits from graduate students and visiting faculty on sabbatical terms.

2.3 Other funding sources

Eawag will provide the necessary infrastructure for the workshop (venue, consumables, IT, etc.). No other funds are needed to support this workshop.

3. Partnership aspects

3.1 Complementarity aspects

We have selected a list of participants that represent a wide range of ages, expertise, and levels of experience. We have identified a mixture of people with (i) skills in mathematical biology, (ii) experience with software development, and (iii) strong conceptual skills in evolutionary biology, meta-ecosystem theory, model systems and field studies.

3.2 Potential for further collaboration

This workshop will lead to new international collaborations. We see great potential in

linking the research of the participants to produce new synthesis in merging coevolution, interaction networks and meta-ecosystems. Specifically, deciphering the interplay between coevolution in local ecosystems for antagonistic and mutualistic interactions with the effects of migration under different landscape arrangements is a topic that can move forward the needed synthesis between the ecological and evolutionary processes that shape biodiversity. Furthermore, our understanding of coevolution among many interacting species at different spatial scales and its effects on biodiversity dynamics and ecosystem function is just starting to emerge. Yet, the integration between robust empirical patterns and process-based models, the main aim of this workshop, is currently missing.

Table 1: Empirical data

	Traits (R/C¹)	Space/Habitats/Distance matrix	Species	Reference
Plant-Bird-Reptile	R:No/C:Yes	Yes/Yes/Yes	R:45/C:13	<i>Heleno et al 2012</i> (approx. 5k bird-reptile individual observations)
Plant-Hummingbirds	R:Yes/C:Yes	Yes/Yes(gradient)/Yes	R:40/C:23 Genera	<i>Weinstein and Graham, 2017</i> (approx. 5k hummingbird individual observations)
Plant-Pollinators	R:Yes/C:Yes	Yes/No/Yes	R:96/C:172	<i>Aizen et al., 2016</i> (approx. 6k pollinator individual observations)
Host-virus	R:Yes/C:Yes	No (replicates)/No/No	R:Types/Types	<i>Frickel et al., 2016</i>

1 R: resource trait and C: consumer traits

3.3 References

- Aizen, M. A., Gleiser, G., Sabatino, M., Gilarranz, L. J., Bascompte, J., and Verdú, M. (2016). The phylogenetic structure of plant-pollinator networks increases with habitat size and isolation. *Ecology Letters*, 19:29-36.
- Allhoff, K.T. and Drossel, B. (2013). When do evolutionary food web models generate complex networks? *J. Theor. Biol.*, 334:122–129.
- Althoff, D. M., Segraves, K. A., and Johnson, M. T. J. (2014). Testing for coevolutionary diversification: linking pattern with process. *Trends in Ecology and Evolution*, 29:82-89.
- Astegiano, J., Guimarães Jr, P. R., Cheptou, P. O., Vidal, M. M., Mandai, C. Y., Ashworth, L., and Massol, F. (2015). Persistence of plants and pollinators in the face of habitat loss: insights from trait-based metacommunity models. *Advances in Ecological Research*, 53:201-257.
- Andreazzi, C. S., Guimarães, P. R., and Melián, C. J. (2018) Eco-evolutionary feedbacks promote fluctuating selection and long-term stability of antagonistic networks, *Proceedings of the Royal Society B: Biological science*, 285:20172596.
- Borthagaray, A. I., Barreneche, J. M., Abades, S., & Arim, M. (2014a) Modularity along organism dispersal gradients challenges a prevailing view of abrupt transitions in animal landscape perception. *Ecography*, 37:564-571.
- Borthagaray, A. I., Arim, M., & Marquet, P. (2014b). Inferring species roles in metacommunity structure from species coexistence networks. *Proceedings of the Royal Society B*, 281:20141425.
- Borthagaray, A. I., Pinelli, V., Berazategui, M., Rodriguez-Tricot, L., & Arim, M. (2015a) Effects of metacommunity networks on local community structures: from theoretical predictions to empirical evaluations. *Aquatic functional biodiversity: an ecological and evolutionary perspective*. Academic Press, Cambridge, 75-111.
- Doebeli, M., and Ispolatov, I. (2017). Diversity and coevolutionary dynamics in high-dimensional phenotype spaces. *The American Naturalist*, 189:105-120.
- DeLong, J.P. and Gibert, J.P. (2016) Gillespie eco-evolutionary models (GEMs) reveal the role of heritable trait variation in eco-evolutionary dynamics. *Ecol. Evol.*, 6:935–945.
- Doebeli, M. (2011). Adaptive Diversification. Princeton Univ. Press, Princeton.
- Guimarães, P.R., Jordano, P. & Thompson, J.N. (2011). Evolution and coevolution in mutualistic networks. *Ecology Letters*, 14:877–85.
- Gravel, D., Guichard, F., Loreau, M., & Mouquet, N. (2010). Source and sink dynamics in meta-ecosystems. *Ecology*, 91:2172-2184.
- Gomulkiewicz, R., Thompon, J. N., Holt, R. D., Nuismer, S. L., and Hochberg, M. E. (2000). Hot spots, cold spots, and the geographic mosaic theory of coevolution. *The American Naturalist*, 156:156-174.
- Fisher R. A. (1930). The Genetical Theory of Natural Selection. (Clarendon, Oxford)
- Fussmann, G. F., Loreau, M. and Abrams, P. A. (2007), Eco-evolutionary dynamics of communities and ecosystems. *Functional Ecology*, 21:465-477.
- Frickel, J., Sieber, M., and Becks, L. (2016). Eco-evolutionary dynamics in a coevolving host–virus

system. *Ecology letters*, 19:450-459.

Hairston, N.G., Ellner, S.P., Geber, M.A., Yoshida, T. & Fox, J.A. (2005) Rapid evolution and the convergence of ecological and evolutionary time. *Ecology Letters*, 8:1114–1127.

Heleno, R. H., Olesen, J. M., Nogales, M., Vargas, P., and Traveset, A. (2012). Seed dispersal networks in the Galápagos and the consequences of alien plant invasions. *Proceedings of the Royal Society B*, 280:20122112.

Hembry, D. H., Yoder, J. B., Goodman, K. R. (2014). Coevolution and the diversification of life. *The American Naturalist*, 184:425-438.

Laroche, F., Jarne, P., Perrot, T., & Massol, F. (2016) The evolution of the competition-dispersal trade-off affects α and β -diversity in a heterogeneous metacommunity. *Proceedings of the Royal Society Biological Sciences Series B*, 283:20160548.

Leibold, M.A., Holyoak, M., Mouquet, N., Amarasekare, P., Chase, J. M., Hoopes, M. F., Holt, R. D., Shurin, J. B., Law, R., Tilman, D., Loreau, M., & Gonzalez, A. (2004) The metacommunity concept: a framework for multi-scale community ecology. *Ecology Letters*, 7:601–613.

Loreau, M., Mouquet, N., & Holt, R. D. (2003). Metaecosystems: a theoretical framework for a spatial ecosystem ecology. *Ecology Letters*, 6:673-679.

Loeuille N, Loreau M. (2005) Evolutionary emergence of size-structured food webs. *Proc. Natl. Acad. Sci. USA*, 102:5761-5766.

Logue, J. B., Mouquet, N., Peter, H., Hillebrand, H., & Metacommunity Working Group. (2011) Empirical approaches to metacommunities: a review and comparison with theory. *Trends in ecology & evolution*, 26:482-491.

Massol, F., Altermatt, F., Gounand, I., Gravel, D., Leibold, M., & Mouquet, N. (2017a) How life-history traits affect ecosystem properties: effects of dispersal in metaecosystems. *Oikos*, 126:532-546.

Massol, F., Dubart, M., Calcagno, V., Cazelles, K., Jacquet, C., Kéfi, S., & Gravel, D. (2017b) Island biogeography of food webs. *Advances in Ecological Research*, 56:183-262.

Melián, C. J., Matthews, B., de Andreazzi, C. S., Rodríguez, J. P., Harmon, L. H., and Fortuna, M. A. (2018). Deciphering the interdependence between ecological and evolutionary networks. *Trends in Ecology & Evolution (In Press)*.

Mouquet, N. & Loreau, M. (2003). Community patterns in source-sink metacommunities. *American Naturalist*, 162:544–557.

Nuismer, S., Jordano, P. and Bascompte, J. (2013). Coevolution and the architecture of mutualistic networks. *Evolution*, 67:338–354.

Pomiankowski, A., and Iwasa, Y. (1998). Runaway ornament diversity caused by Fisherian sexual selection. *Proc. Natl. Acad. Sci. USA*, 95:5106-5111.

Schoener, T.W. (2011) The newest synthesis: understanding the interplay of evolutionary and ecological dynamics. *Science*, 331,426–429.

Thompson, J. N. (1994). The coevolutionary process. University of Chicago Press, Chicago.

Thompson, J. N. (2013). *Relentless Evolution*. University of Chicago Press. Chicago.

Urban, M. C., & Skelly, D. K. (2006) Evolving metacommunities: toward an evolutionary perspective on metacommunities. ***Ecology***, 87:1616-1626.

Wade, M. J. (2007). The co-evolutionary genetics of ecological communities. ***Nature reviews***, 8:185-195.

Weinstein, B. G., and Graham, C. H. (2017). Persistent bill and corolla matching despite shifting temporal resources in tropical hummingbird-plant interactions. ***Ecology Letters***, 20:326-335.

Winegardner, A. K., Jones, B. K., Ng, I. S., Siqueira, T., & Cottenie, K. (2012) The terminology of metacommunity ecology. ***Trends in Ecology & Evolution***, 27:253-254.